

Scalable Inter-County Food Flow Prediction Using Graph Neural Networks

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Abstract

Understanding the spatial distribution of food flows is critical for assessing commodity supply chain resilience, infrastructure planning, and environmental impact. However, high-resolution trade data, especially at the county level and across specific commodity groups, are rarely available. In this study, we present a deep learning framework based on a Graph Neural Network (GNN) to predict food flows at the US county level across seven major commodity categories (Standard Classification of Transported Goods (SCTG) 01-07). Our approach trains GNN models solely on the coarser zone-level data of the Freight Analysis Framework (FAF), leveraging node attributes and inter-regional connectivity to enable cross-scale generalization. Each commodity category is modeled independently and the predictions are evaluated against the FAF5 county-level benchmarks. To support spatial generalization, our model integrates detailed node attributes with enriched edge features, including gravity-based interaction terms, distance transformations, and port infrastructure indicators. The GNNs achieve an average classification accuracy of 0.724 and an average Common Part of Flows (CPF) score of 0.22 at the FAF zone level, effectively capturing structural characteristics of US food distribution. Furthermore, the same models are applied to predict county-level flows, providing scalable and trustworthy estimations that outperform traditional models by over 200% improvement in CPF and generalize across spatial scales without requiring county-level training data. By producing spatially explicit commodity-level flow estimates from coarse training data, our method offers a generalizable and data-efficient tool for applications in supply chain vulnerability analysis, environmental footprint modeling, and regional food system planning.

CCS Concepts

• **Applied computing** → **Agriculture**; Transportation; • **Mathematics of computing** → *Graph algorithms*; • **Computing methodologies** → *Machine learning*.

Keywords

Food flow modeling, Graph neural networks, Freight Analysis Framework, GeoAI, Machine learning, Spatial networks

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1 Introduction

Global food security relies on stable food flows, which are the ways food is produced, processed, distributed, and consumed [5, 8]. Predicting food flows between countries, states, commodity survey zones, and counties remains a significant challenge due to the complexity and heterogeneity of spatial data. Each spatial unit, whether a state, FAF (Freight Analysis Framework) zone, or county, often contains region-specific information that cannot be easily generalized or transferred across geographic scales. For example, while state-level flow data may capture broad trade patterns, they lack the spatial resolution needed to accurately describe county-level dynamics for specific food categories. Conducting fine-grained surveys at the county level or finer resolution is logistically and financially prohibitive, making high-resolution ground truth scarce. Figure 1 shows the aggregated food flows estimates for the FAF zones and US counties by the Bureau of Transportation Statistics, highlighting a spatial distribution pattern in which flows originate from major agricultural production areas and concentrate toward densely populated consumption regions. At the same time, the county-level map reveals finer spatial heterogeneity and more localized hubs that are not visible in the coarser FAF-zone aggregation.

Over the past decades, researchers have developed methods to model subnational food flows using linear programming and mass balance constraints. For example, Lin et al. [11] introduced the Food Flow Model, a data-driven approach that downscales FAF-level flows to counties using supervised learning, production statistics, and distance minimization. Karakoc et al. [7] then extended this model to estimate the temporal dynamics of county-level flows between 2007 and 2017, highlighting key nodes in the national food supply chain through a centrality-based ranking. However, these models are static and rely on strong assumptions such as fixed mass balance and simplified transportation costs. Moreover, they are not easily generalizable across time periods or missing-data settings, and they do not fully utilize the relational geography inherent in the U.S. food system.

Recent advances in graph neural networks (GNNs) offer a promising alternative using deep learning. GNNs have shown strong performance in tasks that involve spatial, relational, and infrastructure networks, learning from both node characteristics and the structure of connections [23, 25, 26]. For transportation and trade networks,



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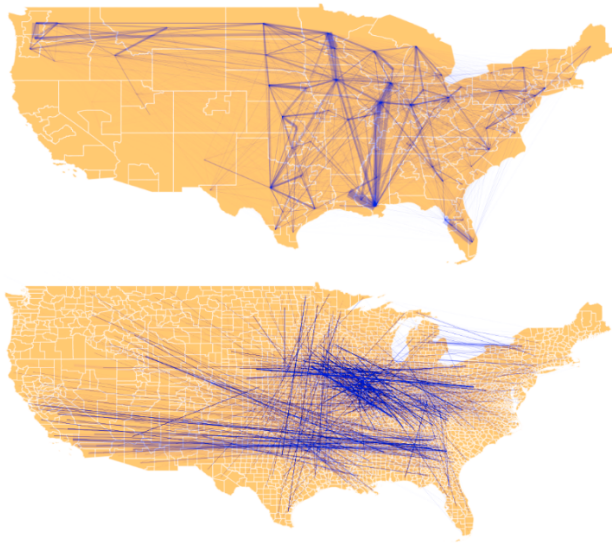


Figure 1: Food flows between the Freight Analysis Framework (FAF) Zones (Top) in Continuous USA for the Standard Classification of Transported Goods (SCTG) 01 to 07 and Inter County Food Flows (Bottom) for SCTG 01 to 09.

GNNs can learn to generalize patterns of flow between connected regions, enabling scalable predictions and improved handling of sparsity. However, the application of GNNs to subnational food trade remains largely unexplored.

Thus, this study aims to contribute to three key perspectives on food flow prediction.

- We examine whether a GNN trained solely on coarse-resolution FAF zone-level data can accurately generalize to predict fine-grained county-level flows, all within the same year. This setting enables us to assess the spatial transferability of graph-based deep learning without the added complexity of temporal extrapolation.
- We evaluate the effectiveness of GNNs in capturing not only flow magnitudes, but also the broader structural organization of the U.S. food distribution network, including regional trade connectivity, spatial heterogeneity, and the hierarchical relationships between production and consumption areas.
- We compare the performance of the model in seven major agricultural and food commodity groups (SCTG categories 01-07), each exhibiting different spatial and economic dynamics, to assess the robustness and adaptability of the GNN framework in commodity-specific trade behaviors.

By training GNN models at the FAF level and applying them to a denser county-level graph, we aim to test the hypothesis that food trade patterns are learnable across spatial resolutions via topological regularities and node-level features, where topology refers to the connectivity of the food flow graph with nodes as spatial units and edges as trade links. Our evaluation compares the predicted flows with the county-level estimates of the FAF (version 5)¹ for

¹<https://faf.ornl.gov/faf5/>

the year 2022, which serves as a latestest version for standardized reference. The resulting framework offers a scalable, data-driven alternative to existing regression or linear programming-based models, with promising implications for the monitoring of the national food system, spatial policy targeting and general-purpose flow estimation in data-sparse contexts. This framework is designed not only for academic modeling but also for practical use. County-level flow predictions can inform infrastructure planning, food supply chain risk analysis, and disaster response, especially in regions that lack detailed food trade data.

2 Related Works

As we mentioned earlier, estimating food flows with high spatial resolution remains a critical challenge to understand supply chain dynamics, regional food resilience, and environmental impacts. Food flows have traditionally been aggregated at FAF zone level, according to the regional functions of commodity flows, leaving a gap at finer scales such as counties. To address this, Lin et al. [11] introduce the first Food Flow Model, a pioneering data-driven framework that uses mass balance constraints and linear programming to disaggregate FAF level flows into county level. This approach considers statistical production and consumption data from each origin and destination pair and forms multiple algorithms related to each different food category. Building on this foundation, Karakoc et al. [7] enhanced the model with tuning, distance smoothing, and extended the predicted results to 2007, 2012, and 2017. These models provided some of the first national-scale, spatially explicit estimates of county-to-county food flows, enabling subsequent research on flow resilience and regional vulnerabilities. However, their heavy reliance on accurate input data and strong structural assumptions limits their flexibility, whereas neural network models, particularly GNN, can generalize more easily across spatial scales and learn relational patterns from incomplete or heterogeneous data.

Previous studies have used simulation and optimization methods to improve food distribution. For example, Validi et al. [22] designed a multi-objective supply chain model, and Birkmaier et al. [1] applied data-driven strategies to reduce loss in perishable food logistics. Building on these efforts, more recent research has turned to network-based approaches to better understand the structure and resilience of food systems. Karakoc and Konar [6] used centrality measures to explore resilience in global food trade, Miller et al. [12] applied network analysis to perishable food movements, while Zhang and Zhou [24] studied crop trade networks using node-ranking methods. These studies highlight the structural complexity of food flows and point to the need for graph-based models that can capture both spatial and relational patterns. Recent advances in machine learning and deep learning have further supported this direction by making it easier to model complex systems like food supply networks, even under challenges such as data sparsity and structural uncertainty.

Among these methods, GNNs are particularly well suited to food flow modeling, where nodes represent zones and edges represent flows between them. GNNs can naturally take advantage of the network structure to learn spatial interactions. Qu et al. [14] introduced FLEE-GNN, a federated learning model designed to analyze multi-commodity food flows and improve resilience across

regions. Similarly, Tang et al. [17] applied a context-enhanced heterogeneous GNN to predict food safety risks, showing the model’s potential in agricultural decision-making. Beyond food systems, GNNs have also been used in related domains such as international trade [13] and mobility flow modeling [16], where learning from topological patterns has led to significant improvements in model generalization. These examples show the growing potential of GNNs to address spatial complexity in systems where traditional statistical models often fall short.

3 Methodology

3.1 Datasets

To support US county-level prediction from coarser FAF-level supervision, we constructed a multiscale dataset that aligns FAF zone-level and county-level spatial units through consistent features and topology. All data correspond to 2017 to ensure temporal consistency across inputs and targets. The dataset integrates economic, demographic, agricultural, and infrastructural indicators relevant to food production and consumption. Table 1 shows the food categories using the Standard Classification of Transported Goods (SCTG) schema and Table 2 summarizes the key data sources in FAF zone-level.

Table 1: SCTG 01–07 Food Commodity Categories

Code	Description	Links
01	Live animals and fish	6680
02	Cereal grains (incl. seed)	6766
03	Other agricultural products	11548
04	Animal feed, eggs, honey, etc.	10889
05	Meat, poultry, fish, seafood	11307
06	Milled grains and bakery goods	11384
07	Prepared foods, fats, and oils	14447

3.1.1 FAF Zone-Level Training Data. Primary training data consists of commodity flow volumes (in kilotons) between 132 FAF zones, as reported in the 2017 Freight Analysis Framework (FAF5) dataset. This resulted in 17424 inter- and intra- flows between and within the FAF zones. Food categories are classified under SCTG codes 01-07, representing the major food and agricultural commodities. Figure 2 shows the food flow mass distribution across SCTG categories. We compute node-level characteristics for each FAF zone, including total production volume, consumption proxy (population and income), and economic outputs from The US Department of Agriculture (USDA) and Bureau of Economic Analysis (BEA) Census sources. Edges are defined between zone pairs with recorded flows and are attributed to the SCTG group, the great circle distance, and the flow volume. This forms the basis of the training graph.

3.1.2 County-Level Prediction Graph. To evaluate the spatial generalizability of the model, we construct an analogous graph over 3144 US counties. Each county node inherits the same feature schema used for FAF zones, computed from disaggregated sources where available. For example, crop and livestock production values are drawn from the USDA 2017 Census of Agriculture, while population

Table 2: Summary of Datasets Used in the Food Flow Model

Dataset	Purpose and Description
FAF (v5) [4]	freight flow data (kilotons/\$) by SCTG, origin-destination pairs, and transport mode
Commodity Flow Survey [21]	Establishment-level freight shipment data that underlies FAF flows; includes commodity type, mode, value, and origin-destination details
Population and Income [2, 3]	County-level population and income data for estimating consumption demand
USDA [19]	County-level unprocessed crop and livestock production data
County Business Pattern [18]	Industry classification for matching production to SCTG
Principal Ports [20]	151 major port locations across USA
Shapefiles	FAF, State, county level shapefiles for calculating overlaps between land areas and distance matrix

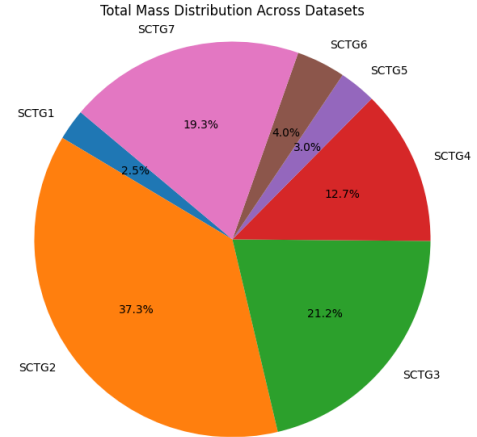


Figure 2: Food Flow Mass Distribution across Categories

and income statistics come from the US Census Bureau. Edge features are similarly defined, with distances calculated between county centroids. In particular, there are no ground truth data on county-to-county flow for 2017; instead, we evaluated our predictions against the modeled county-level estimates provided in the newest 2025 FAF5 release by the US Department of Transportation, Bureau of Transportation Statistics [4]. The estimation provides aggregated results for SCTG code 01 to 07, whereas SCTG 08 (Alcoholic Beverages) and 09 (Tobacco Products) only occupies 5% of the datasets with minimal distraction for evaluation.

3.1.3 Edge Enhancement. In the FAF zone-level data, information such as major mode of transportation, including airfare or truck, is included for each of the flow. However, no similar information is

found for county level predictions. Instead, we use a gravity model based on the population and distance between two locations, to calculate an import-export index in Equation (1) to enrich edge features.

$$F_{od} = G \cdot \frac{P_o \cdot P_d}{D_{od}^\gamma} \quad (1)$$

where:

- F_{od} is the estimated flow from origin o to destination d ,
- P_o is the population at the origin,
- P_d is the population at the destination,
- D_{od} is the distance between origin and destination,
- γ is the distance decay parameter,
- G is a scaling constant.

For this study, we used the distance decay index γ of 2.0 and a scaling constant G of 0.1. For calculating population flows, such an index has been proven to be useful in research [16]. At the same time, the existence of principal ports in each area is defined as a binary feature and will be used as another edge feature. In addition, we include the inverse and log for distance. These approaches aim to enrich the edge level of information, capturing essential spatial frictions and infrastructure hubs that influence trade intensity, particularly in the absence of detailed mode-of-transport data at the county level.

3.1.4 Feature Alignment and Preprocessing. To ensure comparability and transferability, both FAF and county nodes are standardized to share the same 333-dimensional feature vector schema, including all the features from Table 2. Continuous variables are normalized. Principal Component Analysis (PCA) is adapted to reduce feature dimensions to 30. For counties with missing or suppressed values (e.g., due to privacy in USDA reports), we apply imputation using average values. This uniform representation allows for seamless model training at the FAF level and application at the county level, supporting our goal of scalable and cross-resolution prediction.

3.2 Model

Following the approach proposed by Lin et al. [11], as shown in Figure 3, we adopt a hurdle model architecture based on a three-layer Graph Convolutional Network (GCN) or Graph Attention Network (GAT). This model jointly performs feature encoding, binary classification to predict the presence of a food flow link, and regression to estimate the volume of food flows between county or FAF zone pairs. By integrating both tasks into a unified framework, the model captures both the existence and the magnitude of food flows within a graph-based representation of the national food supply network.

We evaluated both the GCN and GAT architectures in order to select the model that yields better predictive performance for each SCTG category. Given that food production and consumption can be driven by inclusive (shared) or isolated (region-specific) factors, different graph structures and attention mechanisms may be more effective depending on the type of product. Therefore, our goal is to identify and apply the most suitable graph deep learning model for each of the seven SCTG categories, enabling tailored predictions that reflect the distinct spatial and economic dynamics underlying each food group.

3.3 Experiments

For each SCTG group, we implemented a GNN to predict county-level food flows, using node and edge features derived from the multiscale dataset. The GNN model consists of a single three-layer architecture with an input feature dimension that matches the size of the node feature vector (333), a hidden layer of 64 channels, and edge feature integration to capture the relational context between pair locations. We trained the model using stochastic gradient descent (SGD) with a learning rate of 0.01 and a momentum of 0.9 to ensure stable convergence. The prediction objective was formulated as a regression task, where the model minimizes the mean squared error (MSE) between the predicted and reference flow volumes (in kilotons). The classification task of identifying whether a food flow link exists between two locations uses Binary Cross Entropy (BCE) loss. All models were trained separately for each SCTG category, allowing category-specific learning while keeping the structure of the training graph consistent between experiments. The dataset is divided into 80% training set and 20% validation set. The dataset is stratified based on percentage of zero links mentioned in Table 1. The training process was efficient; it took approximately 2 minutes for 2000 epochs on CPU, and we chose the best model out of the process.

4 Results

4.1 Primary Results

4.1.1 FAF Zone Level. Although R^2 is a widely used metric to assess the proportion of variance explained by a regression model, it has limitations in the context of spatial interaction modeling, particularly for skewed flow distributions. In county-level food flow prediction, many origin-destination (OD) pairs carry negligible or zero flow, while a few dominate in volume. This imbalance means that R^2 tends to be disproportionately influenced by how well the model fits the largest flows, potentially obscuring its performance in capturing the overall spatial structure. As a result, a high R^2 may not necessarily reflect an accurate modeling of the spatial distribution of flows throughout the network. At the same time, the GCN outperforms the GAT in all categories. A likely explanation is that the relatively homogeneous county/FAF graphs exhibit consistent connectivity patterns, where the added flexibility of attention mechanisms contributes little beyond convolutional aggregation. In this setting, the GCN is better suited to capture broad structural regularities, leading to more stable performance across categories.

$$CPF = \frac{2 \sum_{i,j} \min(X_{ij}, \hat{X}_{ij})}{\sum_{i,j} \hat{X}_{ij} + \sum_{i,j} X_{ij}} \quad (2)$$

In contrast, the Common Part of Flows (CPF) provides a more spatially-aware assessment of the model performance. We adopt a modified version of the Common Part of Commuters (CPC) metric, originally developed in human mobility research to measure the agreement in both the existence and quantity of flows across OD pairs [9, 16]. This study refers to this relationship as the Common Part of Flows (CPF) to reflect its application in food trade modeling. Equation 2, X_{ij} represents the observed flow from origin i to destination j , and $\hat{X}_{ij}$ is the corresponding predicted flow. It is less sensitive to outliers and more reflective of spatial patterns. CPC,

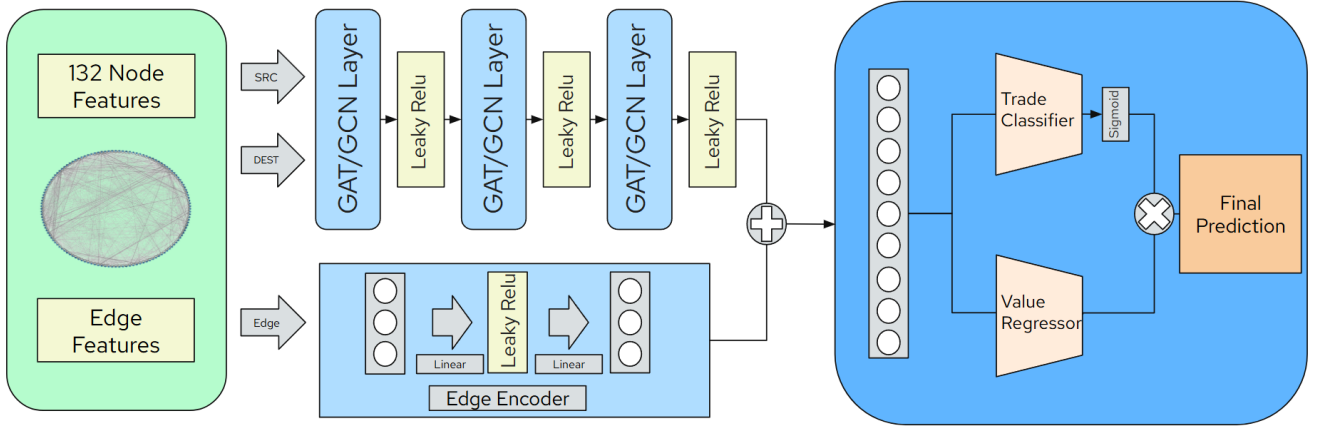


Figure 3: Model Architecture

originally designed for human mobility comparison, measures both the existence and volume overlap of predicted versus benchmark flows. Its value ranges from 0 (no overlap) to 1 (perfect match). CPF is particularly useful for evaluating graph-structured output, as it reflects how well the predicted flow network matches the observed topology. Throughout this paper, we distinguish two forms of CPF:

- **OD-level CPF:** computed mean score across all origin destination pairs, used to assess the model’s global structural fit, and
- **County-level CPF ratio:** computed by aggregating flows for each origin or destination county, used for spatial visualization and localized evaluation.

In this section, we focus on OD-level CPF to measure the overall match quality between predicted and benchmark flows across the national food system. As shown in Table 3, SCTG04 (Animal feed, eggs, honey, etc.) and SCTG07 (prepared foods, fats, and oils) exhibit the two highest CPF values (0.3298 and 0.2412) despite having only moderate R^2 (0.250 and 0.278), respectively.

Table 3: Model Performance by SCTG Commodity Category. Each row reports classification accuracy (ACC), coefficient of determination (R^2), Common Part of Flows (CPF), valid prediction rate (share of OD pairs with positive flow predicted), and the proportional contribution of each category to total food mass.

Category	ACC	R^2	OD-CPF	Non-Zero Rate	Mass
SCTG01	0.778	0.300	0.1908	38.3%	2.5%
SCTG02	0.762	0.240	0.0701	38.8%	37.3%
SCTG03	0.573	0.278	0.1802	66.3%	21.2%
SCTG04	0.601	0.250	0.3298	62.5%	12.7%
SCTG05	0.592	0.140	0.1876	64.9%	3.0%
SCTG06	0.619	0.247	0.2107	65.3%	4.0%
SCTG07	0.580	0.278	0.2412	82.9%	19.3%
Average	0.643	0.248	0.2014	—	—

This contrast indicates the importance of using CPF in combination with R^2 , particularly in food flow modeling, where reproducing the correct spatial topology is often as critical as capturing aggregate variance. Although the model is quite accurate in classification (ACC) in determining if there is any flow between origin-destination (OD) pairs—between 0.573 and 0.778 for SCTG categories, it is misleading to use this metric only in isolation when working with sparse OD matrices, when most OD pairs have zero flow. For instance, SCTG01 (Live Animals) and SCTG05 (Meat, eggs, etc.) exhibit relatively good ACC but have made little input to overall food mass, suggesting that the model performs relatively well with small, isolated flows but poorly with structurally important ones. On the other hand, SCTG02 (cereal grains), the most prevalent category at over 37% total mass, bears a high ACC of 0.762, a low CPF of 0.07 and moderate R^2 of 0.240. This inconsistency is an indication of the lack of strength of the model in accurately capturing the spatial pattern of high-volume flows. More specifically, the low CPF indicates that predicted high flows are spatially non-congruent with benchmark flows, whereas the moderate R^2 suggests a weak fit relative to flows’ magnitudes. Together, these figures suggest the model would tend to under-estimate or mis-allocate bulk commodity flows, often taking multimodal corridors and centralized processing routes not easily derivable from static node-level attributes alone.

Therefore, although high ACC indicates that the model successfully identifies the existence of flows, it fails to accurately identify large-scale food trade magnitude and spatial direction. This limitation emphasizes the need to continue enhancing edge-level representation, e.g., through the incorporation of infrastructure indicators (e.g., ports, intermodal terminals, processing plants) and gravity-based interaction terms, to further guide the location and intensity of large flows in structurally important categories such as SCTG02.

4.1.2 County Level Estimation. Previous results are based on models trained using FAF zone-level data, which provide structured and reliable supervision to learn high-level patterns in commodity movements. As a next step, we compare our deep learning model’s

predictions on county-level flows with the FAF-estimated county-level benchmarks published by the US Bureau of Transportation Statistics [4]. This comparison serves as a test of the model's ability to generalize from coarser (FAF zone) supervision to finer-scale (county-level) prediction.

Theoretically, there are 9,884,736 flows for the inter- and intra-links between 3144 counties for each SCTG category. We aggregated SCTG categories into one number to be compared with the benchmark, which has only the information for the total of SCTG 01 to 09. After filtering out extreme outliers to reduce the influence of anomalously large or small values, we obtain an RMSE of 6.6 and a CPF score of 0.1208. The moderate CPF value reflects a partial but non-trivial alignment in the spatial distribution of flows between predicted and benchmark values. The mean of predicted county-level flows is 4.39 kilotons, which is slightly lower than the FAF-derived benchmark mean of 4.90 kilotons. The previous work from Lin et al. [11] and Karakoc et al. [7] reaches 0.03 and 0.04 for CPF when compared to county-level benchmarks; therefore our model with GNNs achieved about a 200% improvement. The lower CPF in previous models stems from the stark mismatch in flow existence: the original Food Flow Models predicted approximately 160 thousand valid links, while the FAF5 county-level benchmark includes roughly 2.3 million nonzero OD flows. This discrepancy reveals a key limitation in earlier approaches, which tended to produce overly sparse networks due to the constraints of the underlying hurdle and distance-minimizing formulations.

In contrast, our model substantially narrows this gap and improves the CPF in primary results, suggesting improved ability to reconstruct the underlying food trade network topology. Although some magnitude underestimation remains, the enhanced structural alignment supports the feasibility of applying models trained at a coarser spatial resolution to predict food flows at finer granularity with greater spatial fidelity.

4.2 Spatial Distribution of Predicted Food Flows

We begin by examining the spatial distribution of the agreement between predicted and actual food flows using the CPF ratio. As shown in Figure 4, the alignment is spatially heterogeneous across the country and varies between origin and destination counties. Higher destination-side CPF values are found in parts of California, the Southwest, and along the Eastern Seaboard, suggesting that the model captures inflow patterns relatively well in these regions. In contrast, CPF origin values are generally lower and more scattered, with only a few agricultural hotspots, like California's Central Valley, showing moderate overlap. This indicates that while the model learns where food is consumed or processed reasonably well, it struggles more with accurately assigning where the flows originate.

Notably, geographic areas with the highest CPF values often correspond to known hubs of production or consumption, such as Los Angeles County, the New York metropolitan area, and major parts of the Pacific coast. These patterns reflect key trade corridors in the US food system, including west-to-southwest and Midwest-to-east pathways. Overall, the CPF maps highlight both the promise and limitations of the model in capturing spatial flow structure,

and they point to the need for further improvements, especially in modeling origin-side behavior.

To assess how well the model recovers the detailed topology of the food flow network, we analyze the CPF ratio at the county level (Figure 4). CPF measures the overlap between predicted and benchmark flows across all OD pairs involving a given county. Our results reveal a clear geographic asymmetry: origin counties have a mean CPF of 0.120, while destination counties perform better with a mean CPF of 0.183. This disparity reflects structural differences in the predictability of food sources versus food sinks. Whereas production behavior can be decentralized and influenced by regional idiosyncrasies such as weather and market volatility, consumption demand in urban and industrialized regions is more strongly tied to observable features like population and income.

Compared to the Food Flow Model of Lin et al. [11], which relied heavily on mass balance constraints and top-down optimization tied to FAF-level controls, our model is trained solely on coarse regional flows and infers county-level interactions without manual balancing. That we still recover meaningful CPF values, along with a mean CPF of 0.1071 per county pair across all OD links, indicates that the model effectively learns the spatial geometry of food systems and generalizes beyond its training resolution. High-CPF counties tend to align with concentrated agricultural zones and major logistics hubs, further validating the learned structure.

4.3 Edge Enhanced Results

Table 4: Model Performance by SCTG Commodity Category. Each row reports classification accuracy (ACC), coefficient of determination (R^2), Origin Destination-Common Part of Flows (OD-CPF).

Category	ACC	R^2	OD-CPF
SCTG01	0.7893	0.3400	0.2678
SCTG02	0.7733	0.2823	0.1961
SCTG03	0.6582	0.3970	0.1713
SCTG04	0.6824	0.3211	0.1452
SCTG05	0.6883	0.3824	0.2180
SCTG06	0.7182	0.4489	0.2277
SCTG07	0.7601	0.5099	0.2775
Average	0.7243	0.3831	0.2148

4.3.1 FAF Zone Level. Adding edge features significantly improved the model's ability to predict food flows at the FAF zone level. By incorporating gravity model estimates, distance-based metrics, and port information, the model becomes more sensitive to the spatial dynamics that shape real-world trade patterns. This enhancement led to consistent improvements across all evaluation metrics, as shown in the comparison between Table 3 and Table 4.

Most notably, modeled results for several SCTG categories experienced substantial improvement. For example, SCTG01 (live animals and fish) saw its CPF score rise from 0.1908 to 0.2678, an increase of over 40%. This suggests that by adding spatial interaction terms, the model was able to better locate and match the structure of live animal trade, which often follows established agricultural and transportation corridors.

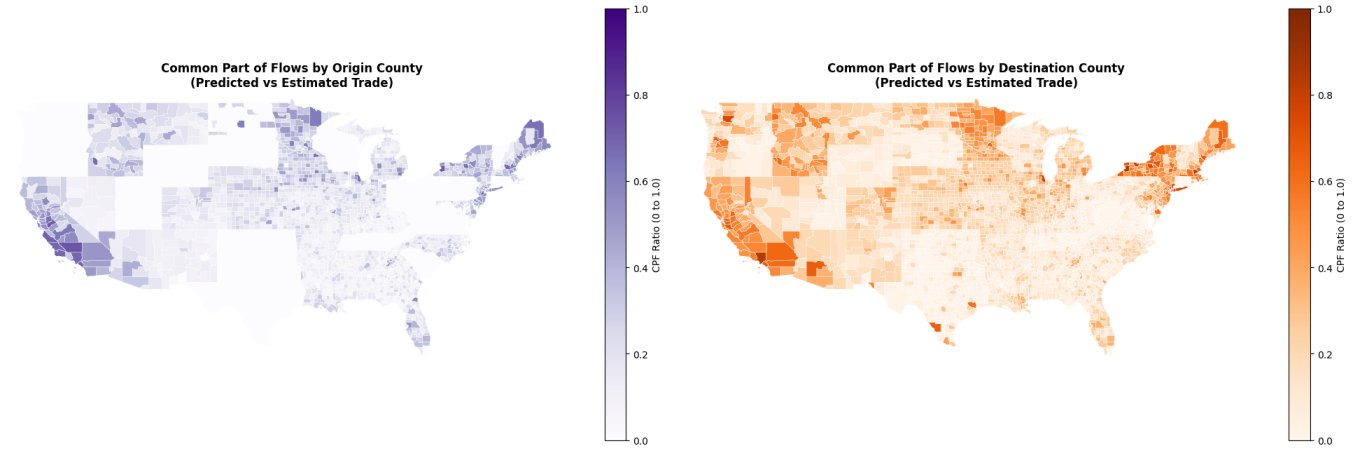


Figure 4: Common Part of Flows(CPF) between estimations and our GNN predictions. Ranges from 0 to 1, darker means more similar. The left panel shows CPF ratios by origin county; the right panel shows CPF ratios by destination county. Destination counties exhibit higher alignment (mean CPF = 0.183) than origin counties (mean CPF = 0.120), indicating stronger model performance in recovering demand-side structure.

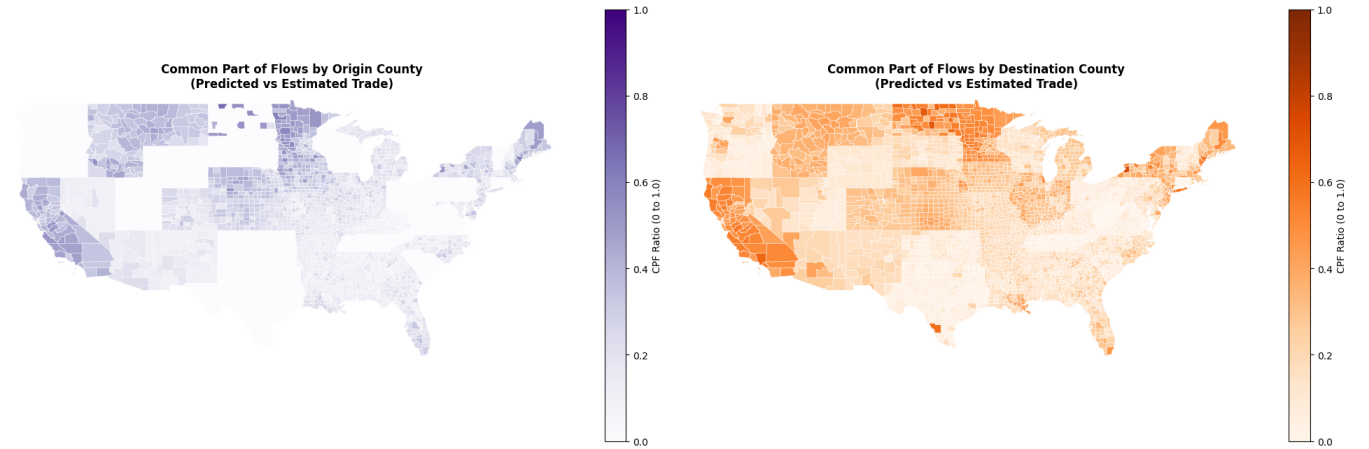


Figure 5: Edge Enhanced Model predicts Destination counties with smoother results than primary results. The Destination County shows higher alignment (mean CPF = 0.1858) than origin counties (mean CPF = 0.1282) again.

SCTG02 (cereal grains), the largest contributor to total flow mass, also showed a marked increase in performance. Its R^2 jumped from 0.240 to 0.2823 and CPF improved from a very low 0.07 to 0.1961. This is especially important because bulk grains move in large volumes along long-distance routes, whereas previously the model had difficulty aligning these flows with ground-truth estimates. The improved results indicate that edge features helped capture not only volumes but also the spatial directions of these critical flows.

SCTG07 (prepared foods, fats, and oils) also benefited from edge enhancement, with its CPF increasing from 0.2412 to 0.2775 and R^2 reaching a high of 0.5099, the best among all categories. This sharp increase reflects the fact that prepared foods often travel through well-defined commercial routes, so that edge features, such as port access or intermodal connectivity, can help model more effectively.

An exception is observed for SCTG03 and SCTG04, where CPF slightly decreases after edge enhancement. These categories aggregate diverse agricultural products and feeds with heterogeneous movement patterns, making them less sensitive to simplified edge proxies such as distance or gravity terms. This suggests that finer-grained commodity breakdowns may be necessary to fully leverage edge-level features.

Overall, we found that edge enhancement strengthened the model's ability to capture both the magnitude and structure of food flows, particularly in key commodity groups like cereal grains, live animals, and prepared foods. These improvements demonstrate the value of enriching edge-level information, even when detailed data on transport mode is unavailable. The results suggest that spatial trade patterns can be better predicted when edge features reflect the economic and geographic frictions that shape them.

4.3.2 County Level Estimation. At the county level, edge enhancements produce a clear improvement in both predictive accuracy and spatial coherence. Although the numerical increase in CPF may appear modest since it moves from 0.120 to 0.1282 for origin counties and from 0.183 to 0.1858 for destination counties. The gain can translate into a noticeably smoother and more consistent flow structure when visualized. As shown in Figure 5, high-CPF counties are no longer isolated outliers. Instead, regions with moderate to strong agreement form broader, contiguous zones, especially along major corridors and population centers.

This visual smoothing is not trivial. In large-scale spatial networks with millions of potential flows, even small improvements in overlap across origin-destination pairs can yield significantly more interpretable patterns. Rather than concentrating predictive accuracy for a few well-known hubs, the enhanced model distributes alignment more evenly, suggesting improved generalization and less overfitting to dominant regions. This is particularly evident in agricultural belts and secondary urban areas, which now appear as more integrated into the national flow structure for food security networks.

Thus, the edge-enhanced model contributes not only to improved numerical performance of the GNN model, but also to a more spatially intelligible and actionable prediction landscape. These qualitative gains support the broader goal of developing food flow models that are scalable, trustworthy and may inform infrastructure planning and food security policy for diverse regions.

4.4 A Case Study Evaluation on Los Angeles's Outflow

Los Angeles County (FIPS: 06037) plays a central role in the US food distribution network. As both a major demand center and a processing hub, it engages in extensive global, local, and regional trade [15]. In the benchmark dataset [7, 11], Los Angeles is ranked as one of the top three counties with the highest total food outflow, reflecting its dual function as a primary origin and destination in national food systems. Our GNN model captures this role well: Los Angeles ranks within the top three in predicted outflows and appears as one of the most connected nodes in the benchmark dataset.

Furthermore, Los Angeles achieves a county CPF score of 0.478. This indicates that nearly half of its predicted flows overlap with those in the benchmark, affirming the model's ability to reproduce the correct spatial distribution of trade relationships. Our model is able to match nine of the county's top ten trading partners identified in the benchmark dataset. The biggest trade partner for Los Angeles County is itself, indicating that major flows occur within self-loops. Counties such as Orange (06059), San Bernardino (06071), Riverside (06065), and San Diego (06073) consistently appear as key partners within California, highlighting a stable and realistic local trade structure. This reflects the dense economic interlinkages within Southern California and the frequent short-range transport of food commodities both into and out of Los Angeles. Similarly, the inflow of Los Angeles County is able to locate the source of inflow efficiently.

However, while structural agreement is strong, the magnitude of food flows predicted by all models vary. For instance, our GNN

model tends to under predict the absolute volume of flows and self-loop activity in Los Angeles. This divergence does not reflect a flaw in the model architecture, rather it highlights the challenge of modeling high-volume trade hubs using purely learned representations. Importantly, our GNN approach achieves a high County CPF score of 0.478, demonstrating that it effectively captures the correct spatial pattern of trade partners and the distribution of flow directions, which is critical when generalized to counties with less observed data.

Unlike traditional linear programming methods [7, 11], which rely on strong assumptions and hard-coded constraints, our model learns commodity trade behavior directly from inter-county features. This flexibility allows the GNN to generalize structural flow patterns across space without requiring commodity-specific tuning or zone-level supervision. The Los Angeles case illustrates this strength: the model correctly identifies key trade corridors and partner counties, even though it was trained without direct supervision at the county scale. Enhancing the model with additional context could further improve volume estimates while preserving scalability and adaptability.

4.5 Food Flow Geovisualization Portal

One major benefit of using a neural network is its ability to predict a far greater number of origin-destination food trade links than traditional models. When learning from node features and relational structure alone, the model identifies both well-established trade corridors and previously underrepresented or unknown connections. For example, GNN is an expanded network of predicted food flows that exhibits great potential to inform decisions on transportation planning, food system resilience, and infrastructure investment.

To make these predictions accessible to a wider audience, we developed an online Food Flow Geovisualization Portal (<https://gnnfoodflowportal.pods.icicleai.tapis.io>), as shown in Figure 6). The portal allows users to explore estimated county-to-county food flows by year and SCTG category. Users can select specific counties to view predicted inflows and outflows, ranked trade partners, and flow magnitudes. Maps and tables provide both regional overviews and detailed records that can support local and national planning. At the same time, users can download the data for their own business or research purposes.

By sharing our results through this interactive mapping platform, we aim to bridge the gap between research and practice. Policy-makers, planners, and practitioners may use the portal to better understand the spatial structure of food movements, identify key logistics nodes, and explore potential vulnerabilities in food supply chains [14, 15].

5 Conclusion and Discussion

This study presents a deep learning approach for estimating inter-county food flows in the United States using graph neural networks trained only on FAF zone level data. Despite no direct supervision for the county scale, the model produces accurate and spatially coherent predictions for seven major commodity categories. This method demonstrates that food trade structures can be learned and transferred at different spatial resolutions using only topological and contextual features.

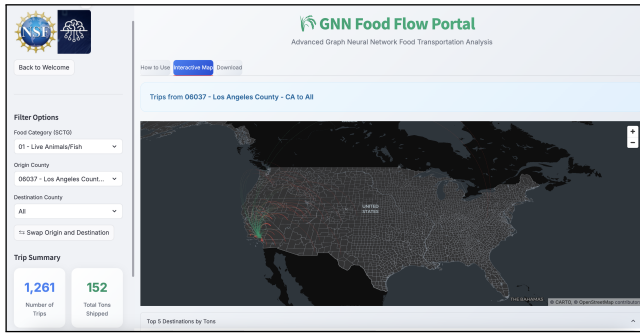


Figure 6: The GNN Food Flow Geovisualization Portal that provides information about flows from origin area to destination area with an interactive map. Users can choose food category, origin, and destination.

First, we validate the spatial transferability of GNNs trained solely on coarse-resolution data. When evaluating against FAF5-derived county-level benchmarks, our model achieves an OD-CPF of 0.22 at the FAF zone level and 0.12–0.18 at the county level, substantially higher than traditional optimization-based models such as Lin et al. [11] and Karakoc et al. [7], which exhibit OD-CPF scores below 0.05 in similar settings. This highlights GNNs’ ability to learn from topological regularities and socioeconomic features to reconstruct fine-scale network patterns at the node level, even without localized supervision.

Second, we find that the GNN approach goes beyond scalar performance metrics such as R^2 by recovering meaningful structural features of food flow networks. The model reproduces known trade corridors, regional inflow-outflow gradients, and self-loops in major food processing hubs. For example, Los Angeles County achieves a county-CPF of 0.478 and a high rank in predicted outflow magnitude, which aligns closely with benchmark data without county-level training. These results confirm that GNNs can not only estimate quantity but also capture the geometry and connectivity of underlying supply chains.

Third, by modeling each SCTG group independently, we show that the framework adapts to distinct, commodity-specific dynamics. Categories such as SCTG07 (prepared foods and oils) and SCTG06 (milled grains and bakery goods) benefit the most from edge feature enhancements, likely due to their dependence on infrastructure and multimodal transport. In contrast, SCTG02 (cereal grains) still poses challenges in magnitude alignment, suggesting that further improvements, including incorporating soft mass balance constraints or multi-scale transport hierarchies, could enhance model realism. Given the large volume of the grain category, breaking it down into five-digit subcategories - wheat, corn, rye, barley, oats, sorghum, and other - may increase the model’s accuracy. Linking live animal movements (SCTG01) with meats (SCTG05) will approximate the end-to-end supply chain and may improve empirical outcomes.

The key advantage of our approach lies in its flexibility and scalability. Traditional disaggregation models rely heavily on hand-tuned assumptions such as minimizing distance, and require repeated calibration for each commodity or year. In contrast, the GNN framework learns from data directly, generalizes across scales, and

seamlessly accommodates both node and edge heterogeneity. This makes our approach especially valuable in data-sparse or rapidly changing contexts, where predefined assumptions may fail.

However, the model still has some limitations. Because it does not enforce mass balance or supply demand constraints, it tends to under predict the total volume in major hubs and may misallocate flows in certain commodity groups. These issues suggest that physical constraints still play an important role in food flow modeling, especially when estimating absolute quantities. At the same time, the benchmark dataset does not represent validated measurements of food volumes, but rather a modeled approximation derived from linear programming and statistical smoothing. As such, discrepancies in magnitude should be interpreted with caution, and our framework is best understood as a complementary, fully data-driven perspective. Although our approach improves CPF by over 200% compared to traditional Food Flow Models, the absolute CPF values (0.12–0.18 at the county level) remain modest. This residual gap suggests that certain structural or economic mechanisms (such as multimodal transport corridors or commodity-specific constraints) are not yet fully captured, and incorporating these elements could further narrow the discrepancy. We also observe systematic differences between predicting origins and destinations: while demand-side patterns in destination counties align more strongly with benchmark data, origin-side flows remain harder to capture. This asymmetry highlights the need for additional features or soft physical constraints to better reflect production-side behavior in future work.

Future work can address these gaps by integrating elements of physical modeling. Soft mass balance constraints could be added to the loss function. Port capacity, transportation mode, and temperature-controlled movements could be used as additional edge features. Combining the learning ability of neural networks with the domain knowledge of traditional models may provide the best of both worlds.

Our results show that GNNs offer a powerful and scalable way to predict food flows. These predictions have direct use for practitioners, planners, food system agencies, and emergency managers. By identifying trade corridors and local bottlenecks, the model supports decisions on investment, logistics, and regional food resilience, even in the absence of detailed ground truth data. Our modeling framework accommodates both node and edge heterogeneity across geographic scales and also provides insights into GeoAI applications on spatial networks [10].

Data and code availability statement

The data and codes that support the findings of this study are available at the following link on GitHub: <https://github.com/GeoDS/GNNFoodFlow/>.

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